Concept Explanation & Memory Analysis

The goal of this task is to compute the attention output without storing the full $n \times n$attention matrix, which leads to high memory usage in standard implementations.

In standard attention, each query vector is compared with all key vectors simultaneously, producing a full attention matrix. This results in **quadratic memory complexity**, making it inefficient for large sequences.

Forward Pass

To reduce memory usage, a **chunked attention approach** is used.

Instead of computing all attention scores at once, the input sequence is divided into smaller chunks of size c. For each query vector, attention is computed only with one chunk at a time.

To ensure correct normalization, **online softmax** is used:

- A running maximum is maintained for numerical stability
- A running sum of exponentials is updated incrementally

The output is computed by accumulating weighted contributions from each chunk. This ensures that:

- the full attention matrix is never stored
- only a small portion of data is held in memory at any time

Despite processing in parts, the final result remains **exactly the same as standard attention**.

Backward Pass

In standard attention, intermediate activations (such as attention scores and softmax outputs) are stored for gradient computation, resulting in high memory usage.

To address this, **gradient checkpointing** is used:

- Intermediate values are not stored during the forward pass
- During the backward pass, required values are recomputed using the same chunked approach

This significantly reduces peak memory usage, at the cost of a small increase in computation.

Memory & Compute Analysis

Standard Attention:

- Memory Complexity: $O(n^2)$
- Compute Complexity: $O(n^2 \cdot d)$
- Requires storing full attention matrix and intermediate activations

Proposed Method:

- Memory Complexity: $O(n \cdot c)$, where $c \ll n$
- Only chunk-sized data is stored at any time
- Compute Complexity: slightly higher due to recomputation.

Comparison

Aspect	**Standard Attention**	**Proposed Method**
Memory Usage	$O(n^2)$	$O(n \cdot c)$
Compute Cost	$O(n^2 \cdot d)$	Slightly higher
Scalability	Limited	Improved
Accuracy	Exact	Exact

Conclusion

By combining chunked computation, online softmax, and gradient checkpointing, the proposed method computes exact attention while significantly reducing memory usage. This makes it suitable for memory-constrained environments without sacrificing correctness.